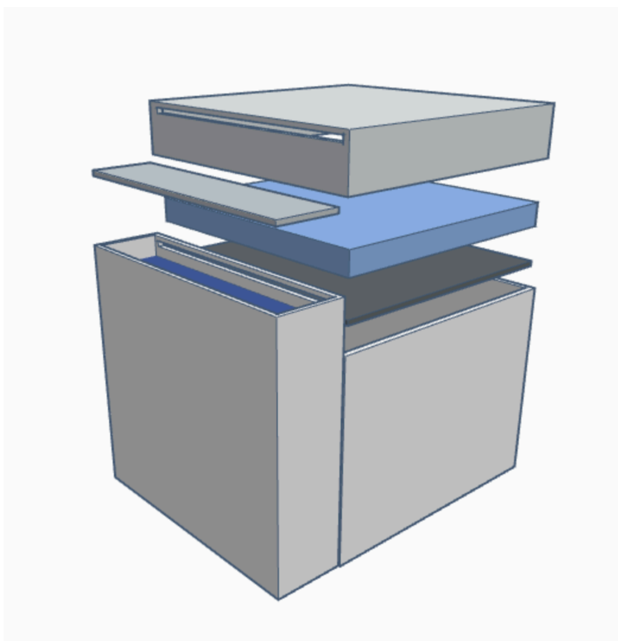
The ZeoBox
Transportable Evaporative Cooling Technology To Reduce Food Waste in India



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Introduction: What is the ZeoBox?

The ZeoBox is an electricity-free refrigeration unit aimed to be used by small farmers in India. It is specifically designed to help resolve the issue of hunger due to the massive amounts of food waste in India. One of the biggest problems facing India's food industry is not the availability or means to make food, it is that a significant portion of the food that is produced is subsequently wasted due to poor refrigeration. According to the UN, 40% of food produced in India goes to waste. This loss occurs at every level of food production including harvesting, processing, packaging, and transporting, to the end stage of consumption. As a result, 194 million Indians go hungry every day (Felder, Steve). This gap is due to issues within the supply chain of farm to table. Our technology aims to increase the availability of food by reducing the amount of food wasted, rather than just increasing food production.

The ZeoBox is designed to be easily transportable so it can be placed on a truck bed or simply used in homes. It utilizes evaporative cooling technology in order to cool up to 50 liters of produce. The ZeoBox consists of a three chamber system.. In one chamber of the box, lies zeolite. Zeolite is a microporous material made of primarily silicon and aluminum. It is a naturally occurring mineral that is created after volcanic rocks come into contact with water. Due to its porous quality, zeolite is most commonly used in an industrial environment to remove metals from water.

The ZeoBox performs as a three chamber cooling system. It contains three boxes, one for produce, another for a water-sand mixture and a

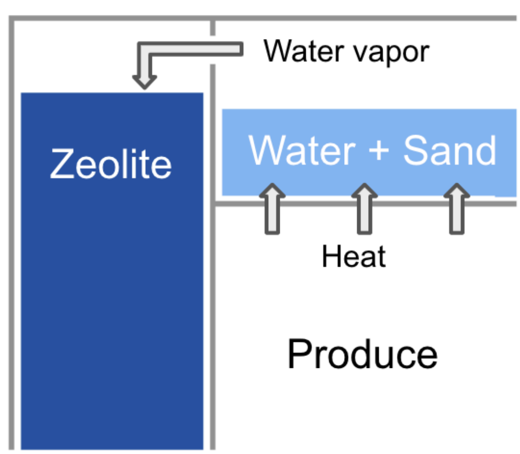


Figure 1

third for zeolite. As a dessicant, zeolite naturally absorbs the water vapor produced from the sand chamber. When the water evaporates into the zeolite, the sand cools down, consequently cooling the produce chamber that sits underneath the sand as shown in figure 1. This process is known as the cooling cycle. The ZeoBox can cool for up to 24 hours, and during this cycle, the zeolite can cool to temperatures as low as 0 ℉. The next step is known as the regenerative process. During this sequence, heat is used to dehydrate the zeolite, restarting the process of the cooling system, allowing for the water to be then reabsorbed by the zeolite and therefore, starting evaporative cooling process again.

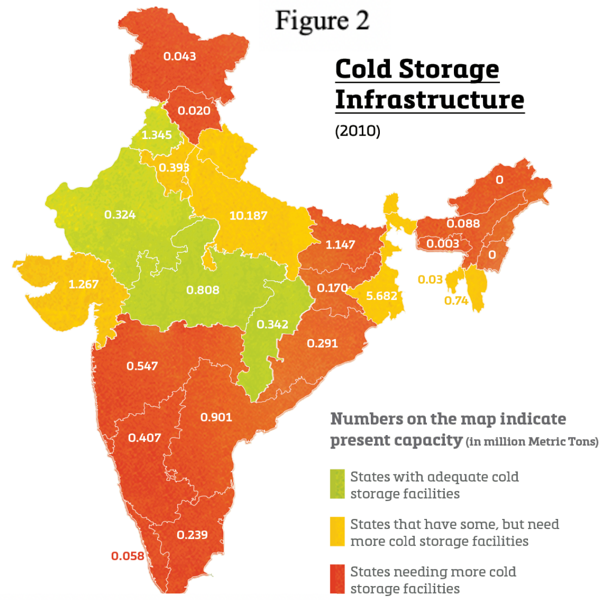
The aspects of this design that set it apart from other cool storage methods are that it can be used in humid climates and that the process does not require any energy source or consistent water supply. It simply provides farmers a place to store their produce in the times when they can not sell their food immediately or they have a surplus of food that should not go to waste.

Target Problem and Population

When assessing the various issues with food production around the world, there were many directions to take our technology. However, we decided to focus on food waste as one third of the food produced in the world is wasted each year. That is approximately 1.3 billion tons of food. Although North America and Oceania wastes the most amount of food per capita consumption, South and Southeast Asia wastes over 115 kg of food per capita each year through the process of agri-production (“Key Facts on Food Loss and Waste You Should Know”). It is much easier to change the systems that govern food production and transportation than the paradigms under which people live their lives and the structures that support this waste of food on the consumer level. Therefore, we decided to focus on the food supply chain, commonly known as the process of farm to table.

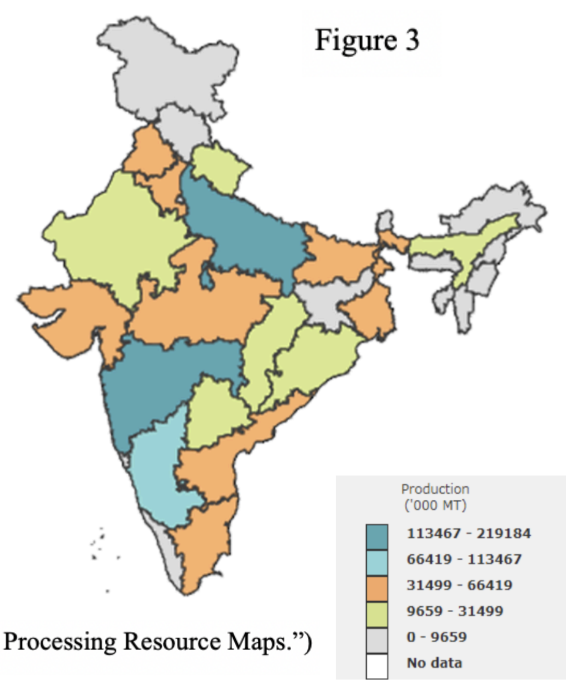
India is a huge supplier of food for the entire world. 14% of global vegetable production is produced by India (Rais, M, and A Sheoran). However, one of the biggest issues with this high capacity of food production is the low quantity of places to store them. There are few cold-storages in India. Of the few that are in India, they can store only 10% of the perishable food, leaving 370 million tons of perishable food at stake (Biswas, Asit K.). Even though India has 6,300 cold storage facilities, 75-80% are suitable only for potatoes, which produce 20% of agricultural revenue (Rais, M, and A Sheoran). That means that 80% of the produce that produce revenue in India are left with only 20% of the cold storage facilities. Figure 2 shows the varying locations of cold storage facilities in India. South and East India is most in need of cold storage infrastructure. Coincidentally, as seen in Figure 3, these are also the areas that produce a majority of the food in India. This fact is especially true in the state of Maharashtra. This Indian state needs cold storage facilities as it can only hold 0.547 million metric tonnes of food even though it produces over

Figure 2



("The Future of Refrigeration in India.")

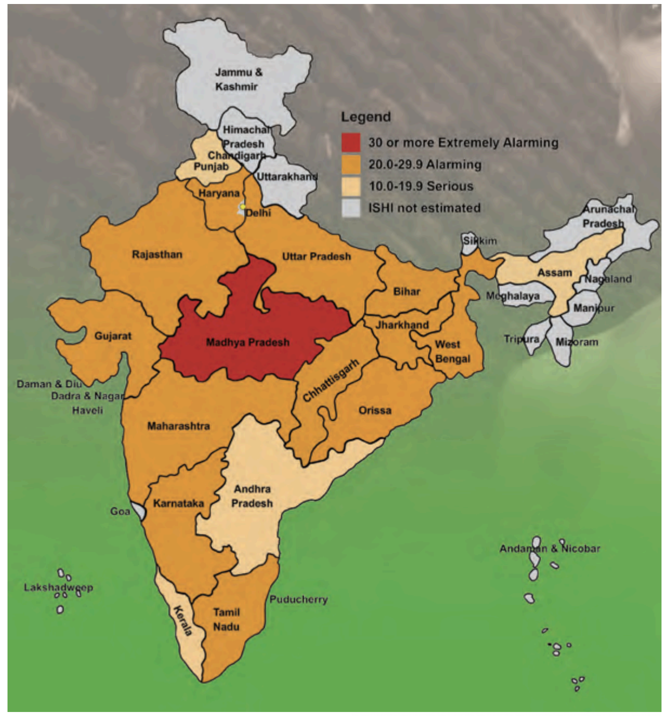
Figure 3



("Food Processing Resource Maps.")

113,467 megatons of food. Consequently, there is a clear need for more affordable and accessible refrigeration technologies.

Figure 4



(Menon, Purnima, et al.)

Despite the fact that India is a supplier of significant amounts of food, 21% of the Indian population is malnourished. Malnourishment commonly results in a loss of productivity, impaired cognitive development, and increased healthcare costs (Thomson, William). A report from world bank found "productivity losses in India due to stunted growth, iodine deficiencies, and iron deficiencies are equivalent to almost 3 percent of GDP" (Thomson, William). The main cause of hunger in India is poor food distribution systems. Figure 4 above shows the graph of the varying global hunger indexes in each state of India. The Global Hunger Index is used globally to track and measure hunger per capita. The state Madhya Pradesh has the highest GHI even though it is between two states that have the highest amount of food production in India (Menon, Purnima, et al.). By juxtaposing these two issues, the importance of creating safe and affordable cold storage solutions for farmers is revealed.

It is established that there are very few cold storage facilities in India. However, small farmers are often cut out of the cold storage facilities available by larger produces in favor of reaching lager markets. This unfortunate fact is true even though these small farmers make up a large portion of the Indian agricultural sector. Over half of the population is a small farmer and two thirds of the same population live in rural areas (Frayer, Lauren). The economic pressure of having good harvests in order to support their family and community has been shown to increase the suicide rate of small farmers in India. India's government data shows that 12,602 farmers killed themselves in 2015 alone due to this financial stress (Changoiwala, Puja). The people of India rely off of the produce cultivated by small farmers. However, without giving access to cold storage facilities to improve the shelf-life of their food, it goes to waste and the people continue to go hungry. Individualized refrigeration systems that could be utilized by these small farmers would be useful in not only allowing them to provide for their family but also opening them up to larger markets in the community through the transportation of their food. Finally, small farmers often have trouble getting access to and using some of the solutions currently available due to their high cost and energy use. According to the book, *Portfolios of the Poor*, many small farmers that live in Southeast Asia have trouble allocating large sums of money to items even though they have a large benefit on their future. Many live their lives on a week to week with a small transaction basis so spending $20 on one item seems irrational (Collins, D.). Therefore, our technology has to address this by keeping costs as low as possible.

Minimizing food waste is the best option in order to solve this problem of hunger in India. By even saving one-fourth of the currently global wasted food, 870 million hungry people in the world could be feed, "about 194.6 million are in India" (Khadka, Shyam). However, this solution demands to be available to small farmers who make up the majority of agricultural production in India and have a large impact on local food availability. The Zeobox was created to target and hopefully relieve some of the hunger, food waste, and economic stress experienced by many people in India.

Existing Technologies

The Zeer Pot

There are many different forms of zeer pots, which keep food cold by using evaporative cooling instead of using electricity. They are used by poor farmers in hot and dry developing countries to make their produce last longer. They can be made out of local materials such as clay and sand. A zeer pot consists of one clay pot places inside a larger clay pot, as seen in Figure 5. Then, sand and water is added to the space between the two pots and the top is covered with a wet cloth. Water needs to be added to the zeer pot approximately two times a day to to keep the sand wet. In addition, it needs to be placed in a dry area with air flow to help aid in the evaporative cooling process. The evaporation cools the inner clay bowl that hold the produce by 10 to 15 degrees Celsius. This product can keep crops fresh for three to four weeks depending on the type of produce. (“Zeer Pot Fridge.”)

Since zeer pots can be handmade they range in size, price, and performance. They can hold as little as 10 liters to as much as 150 litres using anywhere between 1.5 and 2.5 liters of water a day. In addition, depending on the size and material, zeer pots can range in time they take to cool and how long they stay cold for. Similarly, their cost ranges from $2 to $20. They last until they are damaged or broken. Even though the zeer pot is a simple solution it cannot be used in regions that has as temperature below 25 degrees Celsius and that have a humidity greater than 40%. As a result, they can not be used by all poor farmers. (“Zeer Pot.”)

Figure 5 (“Zeer Pot Fridge.”)

<u>Evaptainer</u>

In their attempt to combat this problem of food waste in developing countries, students at Massachusetts Institute of Technology worked to develop a simple, efficient, electricity free refrigeration unit called the Evaptainer, seen below in Figure 6. Their goal is to help small rural farmers in areas such as Sub Sarah Africa, that do not have access to electricity, to increase the amount of crops they get to sell. Their most recent model, the EV-8, uses evaporative cooling to reduce the temperature of its contents by 15 to 20 degrees Celsius. The EV-8 has an inner rubber tub, a layer of water, an outer layer made from their patented PhaseTek™ technology, and is insulated with expanded polypropylene. The PhaseTek™ technology is semi permeable allowing for more efficient evaporative cooling. The Evaptainer holds 1.5 liters of water and uses about 1 liter of water a day. Furthermore, it is durable, lightweight, and portable. (“Evaptainers.”)

Although the Evaptainer provides dependable refrigeration, it has its flaws. Evaporative cooling is less efficient in humid climates. Thus, the Evaptainer does not work in humid climates, limiting is rage of areas it can be used. More specifically, once the humidity reaches above 40%, the Evaptainer will not work. In addition, the Evaptainer is on the market for about $25, which is expensive for their targeted population of poor farmers. It is currently being piloted in Morocco. (Caruso)

Figure 6 (“Evaptainers.”)

Zeo-Tech's Zeolite Transport Box

A German company, Zeo-Tech, created a energy efficient transportable box that uses evaporative cooling through zeolite. Their goal is to be able to use this box to transport frozen goods using less energy. Their technology holds 50 liters of produce, and is constructed out of an insulated box that is surrounded by water and is attached to a chamber of zeolite and a heating unit. The zeolite absorbs the water through the process of evaporative cooling, thus cooling its contents to -18 degrees Celsius for approximately 24 hours. The technology takes 30 minutes to cool and two hours of heating to regenerate and then five hours to cool down. Although these unit are more sustainable, they do require electricity and have a lengthy regeneration process (BINE Information Service).

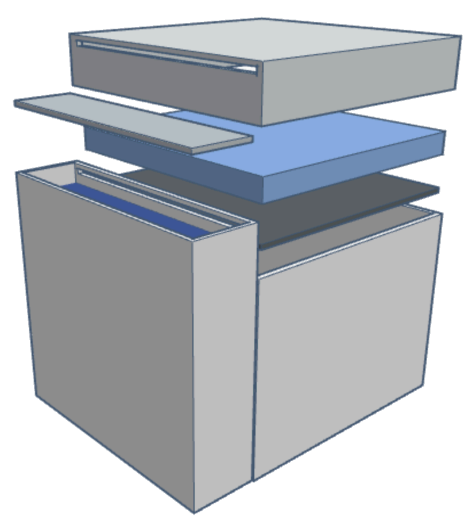
Comparison to ZeoBox

These existing technologies are all created to combat food waste by providing efficient refrigeration. The zeer pot is simple but cannot be used in humid climates. Similarly, the EV-8 Evaptainer is simple, but expensive and it cannot be used in humid climates. On the other hand, the ZeoBox uses zeolite so it does not require a dry climate, so it can be used in humid regions. Thus, our technology is able to cover more regions and be used by more farmers. In addition, our technology is made to be used for transportation, something a zeer pot is not suitable for due to its fragel nature and weight. Zeo-Tech's zeolite transport box is used for transportation and does not require a dry climate, but is does need to be powered by electricity. Therefore, it can not be used by poor farmers who do not have access to electricity. However, the ZeoBox does not use electricity, making it suitable for poor farmers and making it more sustainable. The ZeoBox can be used in humid climes, is cheaper than the EV-8, and does not require electricity, making it better suited for combating food waste especially in poor countries.

Technology Specifications

The ZeoBox is made up of three chambers, one to hold the zeolite, one for the sand and water mixture and one for the produce that is to be cooled. The ZeoBox does not provide the sand or water; users are expected to find these materials in their environments. The technology relies on the process of evaporative cooling using zeolite, which draws in and absorbs water molecules in the air, trapping the

Figure 7



water in its porous material. This process causes the air to dehumidify and allows more water to evaporate. Evaporative cooling occurs when water molecules absorb heat energy and change states to become water vapor. In this transformation, the material from which the water molecules takes heat, in this case the sand, is left cool due to its loss of heat energy. Once the sand is cool, it draws heat from the produce that lies beneath until the sand is dehydrated.

The ZeoBox produce chamber holds up to 50 liters of food, and is 16” x 16” x 12” . The zeolite chamber is 4” x 16” x 15”, while the container holding the sand is 16” x 16” x 3”. In total, the ZeoBox is 20” x 16” x 15”. Between the sand and zeolite chambers is an opening that begins three inches from the bottom of the sand chamber, leaving 1”x16” of space for the water vapor to leave the sand. 9.14 grams of zeolite is needed in its chamber, and .3 liters of water are needed to complete the cooling cycle. The amount of sand can vary so long as it can absorb the .3 liters.

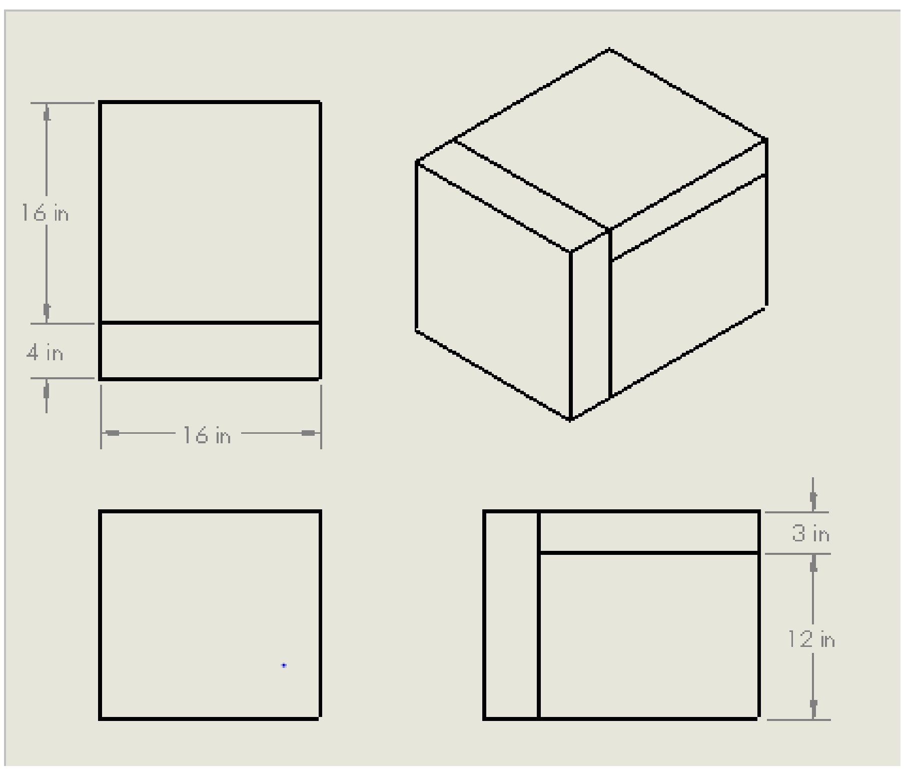


Figure 8: Orthographic Projection

The layer in between the produce and the saturated sand will be made of a carbon steel sheet, as it is a good conductor in addition to being inexpensive. In other evaporative cooling technologies, it was found that the layer around the food itself should be made of a good conductor in order to best transfer the heat from the produce into the sand and water ("Zeer Pot Fridge."). The other sides of the produce box, however, will be made of plastic, specifically PET-G plastic, because it is durable and of low cost. The zeolite chamber will have two layers, one made of PET-G, and another made of carbon steel that fits snugly inside of the plastic box. The inner metal box can be removed by opening the plastic box lid and pulling it upwards. The metal box containing saturated zeolite can be replaced easily with a fresh box of dehydrated zeolite. This design is meant to prolong the amount of time a small farmer may travel with the ZeoBox without having to heat the saturated zeolite containers. Once the zeolite has reached the end of its cooling cycle, approximately 24 hours, the user could just quickly replace the zeolite box to start another cycle, and continue traveling or working until it is more convenient to heat the saturated zeolite box.

The lids of the sand and zeolite chambers are attached to the structure of the box by hinges, and can be opened by pushing them upwards. The produce box can be accessed by pulling down on a specific side attached to the box structure also by a hinge. The plastic sides of the box will be adhered with epoxy, while the metal will be welded together.

Implementation

Selling the ZeoBoxes

The ZeoBox will be primarily used by small farmers in India, both in humid and non-humid climates. Before implementation, the food transportation system in India and how small farmers use this system must be further researched. This includes reaching out to existing organizations focused on helping small farmers in India, such as Kheyti, which implements low-cost technologies for small farmers, and the Smallholder Farming Initiative of Bayer, which aims to help small farmers in developing countries through customized business models. Not only would these organizations give us more insight on lives and needs of local farmers, they could connect us to networks of small farmers, allowing us to interview the farmers directly. After we refine our model based on this research, these organizations could also help us implement our first trial run of the ZeoBox technology using their existing knowledge and networks in farming communities.

Our first trial will consist of giving some ZeoBoxes, for free, to selected farmers both in humid and non-humid climates, and explaining how they work, which is made easy by their simple design. For one year, the usage of the Zeoboxes will be monitored, occasionally interviewing the farmers to see how smoothly they are implemented and find flaws in the design. Then, the technology will be reviewed based on farmer feedback and appropriately modified so that they can be implemented on a larger scale.

Once this trial run is complete, we hope to work with the Indian government to subsidize the technology, using our research during the first trial run to show how this technology could reduce hunger and the amount of farmer suicides nationwide. Then, the ZeoBox will be sold to small farmers, using organizations like Kheyti and the Smallholder Farming Initiative to reach more farmers and educate them about this technology. They will be able to purchase the ZeoBox using payment plans, as their income often fluctuates due to changes in season and often farmers do not hold a large sum of money at any given time. They will also be able to buy as many ZeoBoxes as they need to transport their produce, and farmers in the same community who use a shared truck to transport their produce to a larger market can buy individual units and stack them together on the truck, so that the truck is still shared.

<u>Water Collection</u>

After all of the water has fully evaporated, more water must be collected to refill the units. This responsibility would most likely fall on women, who already take up to six trips a day to gather water in India. These trips can average ten miles a day in rural regions, making them physically exhausting and time consuming. Increasing the amount of water needed would only make it harder for women to get jobs, care for children, and, for younger girls, go to school. (Barton)

There are several solutions which could be implemented to minimize this harmful impact on women and girls in India. While one option is for family members to collect clean water, another option, if there is none readily available, is to use dirty water, which would, for most families, be easier to collect than clean water. The water in the ZeoBox is never in contact with the produce and therefore does not have to be clean. Furthermore, once it evaporates from the sand, it leaves behind any contaminants, both meaning that the sand must be occasionally changed to prevent contaminant buildup but also that the water released by the zeolite during heating is clean, drinkable water. This water is in vapor form, and some sort of additional mechanism would have to be created to catch and condense this water vapor into liquid water. However, this would be relatively simple and inexpensive and would give the ZeoBox a dual purpose: refrigerating produce and filtering water.

If this water condenser is created, another option would be to reuse the water, meaning that water has to be collected much less often. After the first addition of either clean or dirty water is added to the sand, it evaporates into the zeolite, is heated to be released back into water vapor, then condenses and is put back into the sand again, creating a closed-loop system.

More research must be done on water collection in India and potential sources of water which could make the ZeoBox less detrimental to the lives of women. However, as explained above, there are several promising alternatives which could be easily used in the implementation of the ZeoBox.

Heating the Zeolite

After the ZeoBox has been cooling produce for approximately 24 hours, the Zeolite must be removed and heated to release the water. The zeolite chamber can be heated by removing the inner metal layer containing the saturated zeolite from the plastic outer layer. This box, after removing its lid, can then be placed in or next to a fire prepared by the consumers. The zeolite should stay in the fire until it is dried, which could take 2.5 - 6 hours (). It is made out of carbon steel in order to conduct heat well and speed up the heating process. The water in the zeolite then evaporates, dehydrating the mineral, leaving it ready to be reused.

While we decided upon this heating method because it is the most cost effective, there are a few other methods of heating that would require an expansion of this project in order to determine whether or not they are viable. Additionally these other heating options could be a possible additional add-on for farmers to purchase, depending on the situation that they want to utilize the cooling system in. One of these heating options involves using a fresnel lense. A fresnel lense collects and enhances the energy generated by the sun on the Earth's surface. Approximately 1 kW of energy is created by the sun per square meter of the surface of the Earth. This is 0.6 watts per square inch. Furthermore, if you focus the sunlight into an area ⅜ of an inch in diameter, one could create the equivalent of 300 Watts per square inch (Sagendorf, Martin). This energy could be utilized by photovoltaic cells in order to transform the focused sunlight into usable energy to heat the zeolite within its chamber. This would in turn create a closed system which is a huge advantage in this cooling system. A closed system would allow the Zeobox to be reusable without having to constantly clean the system and add more water. This would make it more convenient for transportation services or small farmers that don't have consistent access to heat sources such as a fire since the only thing needed for this system to work is sunlight. While fresnel lense are fairly inexpensive and cost about $3.95, the added prices of other parts of the system like the photovoltaic cells make it less viable as an option for small farmers who have trouble allocating money towards large purchases, nevertheless the impact that they can make on their livelihood ("Fresnel Lenses.").

Cost Estimate

Our initial research into the prices of the various materials that we have planned to use in the creation of our Zeobox estimate the final cost of the technology to be approximately $12.36. The three main materials used in the making of this product is zeolite, carbon steel, and plastic. Figure 9 to the right shows the approximate weights and prices of the three materials used in the Zeobox. Additionally, three hinges, each costing approximately $0.10 will be needed to attached the lids of the chambers (Dayjiuh Interior Hardware).

Material	Amount	Cost per Unit
Zeolite	9.14 kg	$8.22
Carbon Steel	3.41 kg	$2.62
Plastic	10.34 kg	$1.25
Hinges	~0.01 kg	$0.30

Figure 9

Zeolite is the main proponent of the Zeobox. According to Alibaba, the price of purchasing zeolite from Henan Zhongbang Environmental Protection Technology Co. would be approximately $0.90 per kilogram (Henan Zhongbang Environmental Protection Technology Co., Ltd.). Another component of the box is the steel which would be located in two places. It would be at the top of the produce container, separating the produce from the sand and water above it. We decided to use carbon steel because of its efficiency as a conductor of heat to transfer from the produce and into the sand chamber and because in comparison to other metals, it is quite inexpensive when buying in bulk. In order to get the prices of these options, we looked at bulk resources we could purchase from overseas using Alibaba to receive the items. Using the price of carbon steel listed from Tiajin Tsd Steel Imp and Exp Co., $500 per metric ton, we converted the five sheets of carbon steel that we would need into volumes (Tianjin Tsd Steel Imp. & Exp. Co.,Ltd.). In order to convert the dimensions of the sheets of carbon steel into their weight, the density of carbon steel, 7.85 g/cm^3 was used (Elert, Glenn). After we knew the weight of each part that we needed, we used the price per metric ton listed on the site in order to get our final price of each item. The same method was used for plastic and zeolite, except their proper densities, 1.39 g/cm^3 and 1.0 g/cm^3, listed accordingly was used ("Density of Selected Plastic Materials" and "Natural Zeolites & Synthetic Zeolites.").

Plastic was chosen due to its frugality. According to AcmePlastics, the plastic we would need to use, polyethylene terephthalate - glycol, would be approximately $117.99 for 40 sheets at 48" x 96" ("PET-G CLEAR SHEET."). However, the amount of plastic that we would need for a single Zeobox is only $1.25. Plastic was used for the walls of the produce box and double chamber system. However, within the chamber of zeolite is a double layered structure. Inside the plastic walls of the chamber lies a box of carbon steel which is filled with the zeolite. At the top of this compartment contains the crease where water enters from the sand chamber and into this zeolite box. The zeolite is placed in this box in order to be easily taken out and heated by the person utilizing the Zeobox. The total weight of all of these components is 22.89 kg.

Strengths Analysis

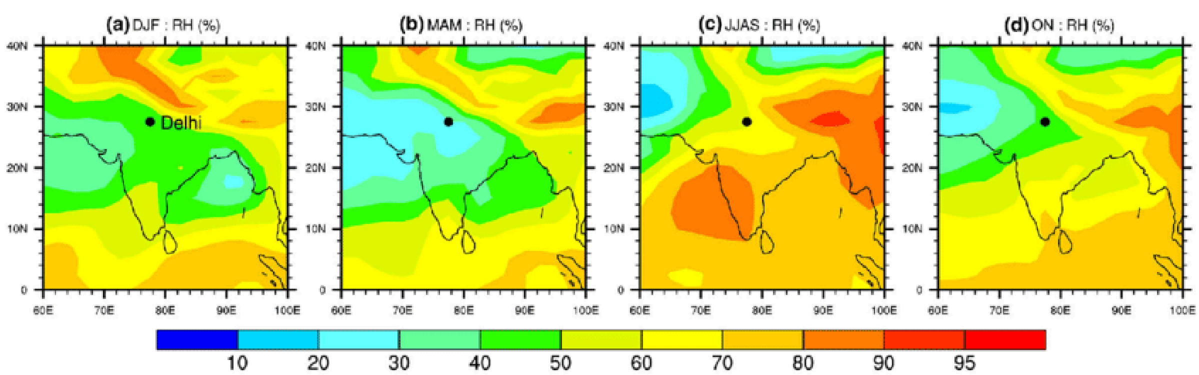
At the beginning of the research process, the zeer pot, Evaptainer, and traditional refrigerating units were all considered as potential modes of transportable refrigeration. All of these technologies, upon further examination, proved to be difficult to apply to the problem of cold chain food transportation in India. The Zeobox was designed with these faults in mind, making it an effective way to refrigerate food cheaply and with little energy use.

Firstly, the ZeoBox targets the heart of the food problem in India. As previously explained, the biggest weakness in the food system is lack of cold chain infrastructure, specifically during food transportation. Compared to increased food production, which requires annual spendings, postharvest facilities require a potentially larger but one-time investment which is compensated for by future savings. Research predicts that a 1% reduction in postharvest losses in India could generate Rs. 230 crores, or approximately half a billion dollars, annually in India. ("Postharvest Management of Fruit and Vegetables") Furthermore, extending the shelf life of produce by just one day has a significant impact on levels of food waste ("What Role Does Refrigeration Play in the Supply Chain?"). The ZeoBox, by lowering food temperatures during transportation, would have a significant impact on food spoilage which would subsequently impact not only hunger rates in India but also the Indian economy. The ZeoBox is also able to function in the humid climates of the southern and western regions of India, where there is even more food wastage due to the tropical climate (Rais and Sheoran). Therefore it would be able to impact the regions where refrigeration is most needed.

Other existing technologies, as covered in the "Existing Technologies" section, also aim to reduce food waste through refrigeration. Traditional refrigeration was the first technology that was considered as a potential solution. However, a refrigerator is heavy, requires a constant source of energy to function, and would be difficult to fix if repairs were needed. The ZeoBox, in comparison, requires no energy during cooling, which eliminates the need to transport a heavy battery. It is also relatively easy to use and fix due to its simplicity, and its stackable design makes it easy to stack into existing trucks and incorporate it into existing food transportation systems in India.

Other cooling mechanisms that were initially researched include the zeer pot and Evaptainer, which also use little to no energy and are cheaper than refrigerator units. However, several issues were found with both technologies. The zeer pot, while cheap, easy to make, and made of local materials, is heavy and fragile, which makes it difficult to transport.The Evaptainer can refrigerate food effectively, however is more expensive. Furthermore, neither technology functions in humid climates, which means that they cannot work in most parts of India. The zeer pot's maximum functional humidity is less than 40% while for the Evaptainer it is 40% ("Zeer Pot.") (Caruso). The primary growing season in India is from June to September, with harvesting in September and October. As shown in Figure 10 (c and d), the average relative humidity from June to September is above 60% for most of India, and from October to November is above 50% for most of India. Consequently, the zeer pot and Evaptainer would not be functional in most of India during the primary harvesting season, when cold-chain food transportation is needed the most.

Figure 10: Graphs of Max Humidity for Zeer Pot



(Bhattacharjee, Parthasarathi, et. al.)

Shows relatively humidity for DJF (December – January – February), MAM (March – April – May), JJAS (June–July–August–September) and OC (October – November)

The ZeoBox itself, like the Evaptainer, was based off of the zeer pot designed but is improved to be transportable and work in humid climates. However it is also not as expensive as the Evaptainer, making it a middle ground between the two that resolves issues in both technologies. It is also lighter and less fragile than a traditional zeer pot, making it more transportable. Zeer pots are also made with two clay pots, making it fragile and difficult to transport, and is not stackable because stacking the zeer pots leads to decreased surface area for evaporative cooling.

Our team predicts, based on our research, that the ZeoBox would work more effectively than the zeer pot and Evaptainer in humid climates. In principle, it is doing the same thing as both technologies: using evaporative cooling to evaporate water into the air, cooling the food. However, one huge drawback in both the zeer pot and Evaptainer is that the water is evaporating into outside air. Both of these technologies work in relatively arid climates, because the air has a greater capacity to hold the evaporated water. In more humid climates, such as most of India, less water is able to evaporate into the air because it is already saturated with water. Therefore, both technologies do not function over a certain humidity level. In the ZeoBox, one important difference is that instead of evaporating into outside air, the water evaporates into air which is essentially dehumidified by the zeolite, making it so more water can be evaporated and more cooling can occur.

Finally, the ZeoBox would be able to help small farmers in India. Because units are small and cheap, farmers will be able to buy as many units as they want depending on their needs and budget. The ZeoBox's stackable design means that it can be attached to existing truck beds, making it easy to implement into the current food transportation system that is used by farmers without any additional infrastructure. Its design is simple and easy to understand and it requires only a source of heat and water, which means it can be implemented in more remote areas. This is particularly important because most of the agricultural activity in India is located in remote areas that do not have existing packaging facilities. (Rais and Sheoran) In communities where multiple farmers transport produce together, they will be able to split the costs by buying their own containers to attach to the same truck. The payment plans for the ZeoBox also make it a more realistic purchase for farmers whose incomes are often seasonal, making it less of a financial burden. If small farmers had access to cooling technology such as the ZeoBox, less of their produce would spoil, leading to higher profits. Furthermore, it would enable them to reach bigger markets because their produce would have a longer shelf life and could be transported farther distances.

All of these factors mean that the ZeoBox could have a significant impact not only on the larger food transportation system in India but more specifically on the lives of small farmers across the country. It would keep produce cool even in humid climates, extending its shelf life so that healthy, unspoiled food can reached those who need it most. It can do so while using minimal amounts of energy and supporting small farmers so that they can make a bigger profit and reach bigger markets. The ZeoBox therefore would have a real impact on the lives of millions, benefitting both the farmers themselves and the millions of Indians that go hungry every day.

Difficulties

Tackling any large-scale problem comes with many challenges, including some that we have already faced and those that have yet to come up during future implementation phases. While in the process of designing our technology, a large obstacle we ran into was the nature of evaporative cooling itself and its limits in humid environments. India is our target country because it is especially lacking in cold chain infrastructure and has unreliable power and electrical sources, causing frequent blackouts. Furthermore, in an article published in The Atlantic a few years ago, it was reported that “more than half of India’s 6,300 cold storage facilities are concentrated in just four of the country’s 29 states,” with these four states being in the northern, more arid regions of India. Thus to target the southern regions of India specifically, we decided to design an electricity-free device using evaporative cooling to bypass this problem of unreliable electricity. However, we did not foresee the problem of the effectiveness of evaporative cooling, which greatly drops in climates of high humidity. This limited the states in India where our technology would be practical to consider to the upper half of India and the more elevated regions. Due to the pressing need for more cooling technology in the majority of coastal and southern regions in India, we decided to pursue ways to alter our technology to address this problem.

We discovered Zeolite, a porous desiccant which works through the process of adsorption to pull water vapor out of the atmosphere. Once we incorporated Zeolite into the design in combination with evaporative cooling, we created a closed container effective at cooling regardless of humidity. So although we had to narrow our region of India where our technology would be effective for a stage of our planning, eventually we were able to expand it to all states with those in most need of local cold chain facilities being our main target.

However, we are still looking further into the challenge of the most feasible and sustainable way to heat the zeolite for regeneration. Currently the distance that the ZeoBox can transport while refrigerating is limited by one complete absorption of the zeolite which is approximately 24 hours, where it would then be necessary to stop and regenerate the zeolite before continuing. Most farmers could create heat in the same way they cook food: by building a fire. However, This poses the challenge of obtaining wood to burn for the fire, especially if farmers are in transit, and could be very labor and time intensive. Fires also contribute to air pollution inside homes when they have poor ventilation. We hope to address this by continuing to research an electricity free heating component to add and potentially collaborate with organizations or companies with the leading low-cost eco friendly stoves or solar powered heating technologies that can be sold along with our ZeoBox to improve its usability. We have continued to upgrade our design with this problem in mind by adding a thin carbon steel sheath that contains the zeolite and can be easily removed from the ZeoBox. The user would be able to heat the zeolite in this sheath eliminating the hassle to handle the zeolite directly or find a container to heat it in.

An important note is that while based on research, our design is still a theoretical technology that has yet to be tested. Therefore our team will still have to create multiple prototypes to more accurately measure the details of our design such as how much zeolite is needed and how efficient the adsorption and heating for regeneration processes really is. There is the possibility that the temperature the Zeobox steadily reaches is not as low as predicted, however, we are well prepared to modify our prototypes based on the results to create the most effective model we can.

Furthermore there are a few technical considerations to keep in mind with our technology. Though we tried to maximize the surface area of water exposed, there had to be some compromise with the size of the container and the effectiveness of the evaporative cooling which is dependent partially on surface area to volume ratio of the container. Also, the amount of produce the farmers can carry in each container is limited by the 50 L size of the current model of the ZeoBox. However, we hope to continually improve on this model and, depending on the feedback of the users of the size of our first design and could create a larger range of options of sizes of the ZeoBox.

We are also implementing this technology on the basis of a few assumptions that could complicate the process. For example, since our technology is not mobile itself, it was created on the assumption that the buyers or farmers already own or at least have access to some sort of transportation such as a truck or similar vehicle. Since the range of our Zeobox's cooling capabilities is currently 24 hours, we are also assuming that there exists at least one other village or market within that span of distance that the produce could be transported to to be sold or stored. Unfortunately, since this is a hard and rigid container, it will not compromise its dimensions to compress into smaller spaces or be easily carried. Therefore, if there are poor roads or areas with paths incapable of driving through, it could cause a geographical complication with using the ZeoBox that the current model is not yet prepared for. We also assume that the farmers are in possession of some type of rope or objects to secure the Zeobox on the vehicle so that it does not slide off in transit.

The ZeoBox also comes with all materials provided except for the sand and water in the upper layer of the ZeoBox which are to be provided by the user from their local area. Though it uses a fairly minimal amount of water with only approximately 0.3 liters needed for each 24 hour cycle of cooling with the zeolite, it could be a potential barrier to buyers if they do not have access to water easily. A positive aspect however is that the water put into the ZeoBox does not have to be clean or drinkable water but can be water from any source, contaminated or not. In addition, once the water is dispelled from the Zeolite, it will be pure water since only pure water vapor is absorbed by the Zeolite while the contaminants remain in the sand, which would likely have to be cleaned out and replaced on a regular basis. However, we still acknowledge the value of water and the possible difficulty of finding that portion of water to allocate to the ZeoBox.

Though we tried to make the technology as low-cost as possible at only $12.06 per container, investing in several containers at once might be too much for a single farmer. Therefore, farmers might either need to invest in one unit at a time and save up to buy multiple when they can afford it, or it could be a joint community effort of multiple farmers pitching in to buy the technology and splitting the profits.

There are also some potential cultural difficulties that we would have to look into further, conducting some studies to smoothly implement our technology and have the farmers adopt the technology into their practical lives. One such example we need to be aware of is the competition it could cause in the village. If those wealthier in some villages could start transporting goods to other village's markets it could create a more competitive market that could make prices for produce more fair, but also cause the sellers to earn less profit. Another is our means of distribution, where further research would be helpful in identifying the best villages to begin distributing in and in working with NGOs or the government on subsidizing the technology for farmers. Distributing it to handpicked families in most need of the Zeobox in a village might also help the technology to be better received by the community.

Despite the potential challenges, we are confident that with further research and prototypes to improve the technology and programs to implement it into villages, the ZeoBox could vastly improve the local cold chain network and trade between regions and reduce the amount of food waste in India.

Conclusion

Grant money would be used to cover the costs of materials and the assembly of the ZeoBoxes as well as the field research that is needed to be done on India's food transportation systems. The distribution and implementation would ideally be subsidized. The grant money would also pay for the produce needed to test ZeoBox's refrigeration abilities as well a vehicle to test its transportability. While zeolite has been safely used in refrigeration technologies in the past, we will conduct further research into the health impacts of prolonged exposure to the mineral using the grant money ("Evaptainers."). The technology design could be further developed by making ZeoBoxes of varying sizes to accommodate ate the needs of different farmers and homes. Another aspect of the technology open to development is incorporating a system that traps the clean water produced in the evaporation process so that families and farmers could have greater access to clean, drinkable water. Some of the grant money will be allotted to developing these components.

The initial steps in the research and prototyping process will be contacting the makerspace of the Johns Hopkins University, FastForward U, to use its facilities and tools. Further steps include a trial run in India to see what design components need to be changed and enhanced to not only be more functional, but more accepted within the small farmer community.

The ZeoBox has the potential to provide cold chain infrastructure to transport produce, significantly reducing food spoilage. Consequently, more people within India will have access to food, reducing national hunger rates. Additionally, besides food transport, the ZeoBox is also suited to transport vaccines which require stable cooling conditions to less accessible regions of India. It can work in hot and humid climates where it has been difficult to find an affordable method of refrigeration. Not only is it affordable, but it also requires less energy, making the ZeoBox a potential long-term solution to the problem of refrigeration and hunger in India.

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